

Study of the out-of-equilibrium dynamics of one-dimensional Bose gases

Étude de la dynamique hors équilibre des gaz de Bosons unidimensionnels

04-11-2025

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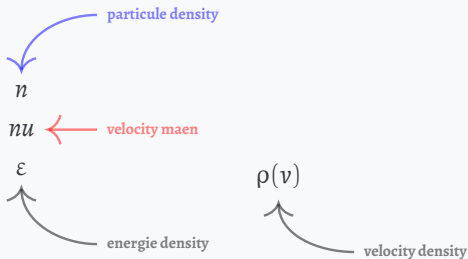
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Rapidities and GGE in a classical system

Chaotic/Ergodic

Integrable/non-ergodic



$\xRightarrow{\text{thermalisation}}$

Thesis Problematic

Scientific Question

How can we characterize the spatio-temporal evolution of a 1D quantum gas after a perturbation, in terms of the local rapidity distribution and its link to Generalized Gibbs Ensembles (GGE)?

One-dimensional quantum gases provide an ideal platform to study strong correlations and non-ergodic dynamics in out-of-equilibrium systems.

Despite major progress, understanding the local rapidity distribution remains a key challenge.

or Schematic view : from experimental system to theoretical description.

Objectives

- Lieb and Liniger;
- GHD;
- Manip;
- Bipartition;
- Fluctuation

The N-body problem: the scattering phase

Contact potential

$$\hat{H} = -\frac{\hbar^2}{2m} \sum_{a=1}^N \hat{\partial}_{x_a}^2 + g \sum_{1 \leq a < b \leq N} \delta(\hat{x}_a - \hat{x}_b)$$

$$\psi_{\{\theta_a\}}(x_1, \dots, x_N) \propto \sum_{\text{perm. } \sigma} e^{i\Phi_\sigma(\{\theta_a\})} e^{i\frac{m}{\hbar} \sum_{b=1}^N \theta_{\sigma(b)} x_b}$$

$$\frac{m}{\hbar} \langle \theta_a \rangle L + \sum_{b=1}^N \Phi(\theta_a - \theta_b) = 2\pi I_a \quad a = 1, \dots, N$$

figures/Figures.pdf

$L\rho(\theta)\delta\theta$	: #	rapidities	$\in [\theta, \theta + \delta\theta]$
$L\rho_h(\theta)\delta\theta$	: #	holes	$\in [\theta, \theta + \delta\theta]$
$L\rho_s(\theta)\delta\theta$	: #	states	$\in [\theta, \theta + \delta\theta]$

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Comparaison des commandes

- **Uncover** : espace réservé

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Subtitle here

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- Dolor sit amet
 - Consectetur
 - Adipiscing elit
- Sed do eiusmod
 - Tempor incididunt
 - Ut labore et dolore
 - Magna aliqua

A frame with title only

Theorem

$$e^{i\pi} + 1 = 0$$

Démonstration

$$e^{iz} = \cos z + i \sin z$$

therefore

$$\begin{aligned} e^{i\pi} + 1 &= \cos \pi + i \sin \pi + 1 \\ &= -1 + i \times 0 + 1 \\ &= 0 \end{aligned}$$

A plain frame has no headline

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<code>\AlegreyaMedium</code>	Lorem ipsum dolor sit amet
<code>\AlegreyaExtraBold</code>	Lorem ipsum dolor sit amet
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Alert! A *plain* frame does not show the progress bar but still appears in it unless the frame comes after `\End`

A ***STANDOUT*** frame can be used to focus attention

In combination with *plain*,
it makes a nice thank-you slide!



<https://github.com/piazzai/arguelles>
<https://ctan.org/pkg/beamertheme-arguelles>

